

# Bitwise differential trail through **TWINE** found by SAT

CiVerLy

July 24, 2026

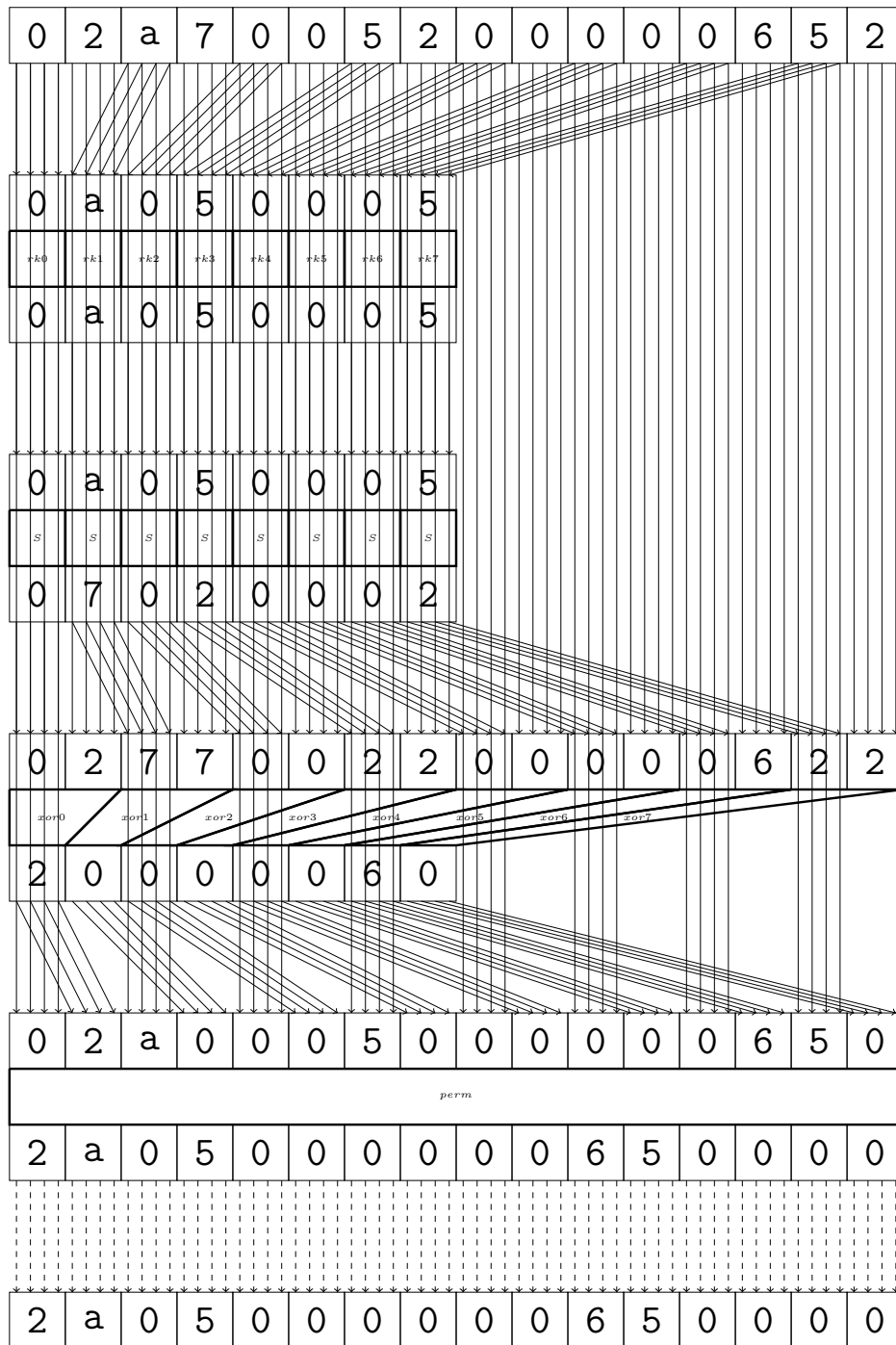
# 1 TWINE

Maximal differential probability:  $2^{-22}$

|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0     | 2 | a | 7 | 0 | 0 | 5 | 2 | 0 | 0 | 0 | 0 | 0 | 6 | 5 | 2 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 2 | a | 7 | 0 | 0 | 5 | 2 | 0 | 0 | 0 | 0 | 0 | 6 | 5 | 2 |
| round |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2     | a | 0 | 5 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 6 | 5 | 0 | 0 | 0 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2     | a | 0 | 5 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 6 | 5 | 0 | 0 | 0 |
| round |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 0 | 5 | 2 | 0 | 0 | 0 | 6 | 0 | 0 | 0 | 0 | 0 | 0 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 0 | 5 | 2 | 0 | 0 | 0 | 6 | 0 | 0 | 0 | 0 | 0 | 0 |
| round |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 0 | 0 | 0 | 6 | 5 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 0 | 0 | 0 | 6 | 5 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| round |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 6 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 6 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| round |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 0 | 6 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 0 | 6 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| round |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 0 | 0 | 0 | 0 | 6 | 0 | 0 | 0 | 0 | 5 | 0 | 0 | 0 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 0 | 0 | 0 | 0 | 6 | 0 | 0 | 0 | 0 | 5 | 0 | 0 | 0 |
| round |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 6 | 0 | 2 | 0 | 0 | 0 | 0 | 5 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 6 | 0 | 2 | 0 | 0 | 0 | 0 | 5 |
| final |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 6 | 5 | 2 | a | 0 | 0 | 0 | 5 |
|       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0     | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 6 | 5 | 2 | a | 0 | 0 | 0 | 5 |

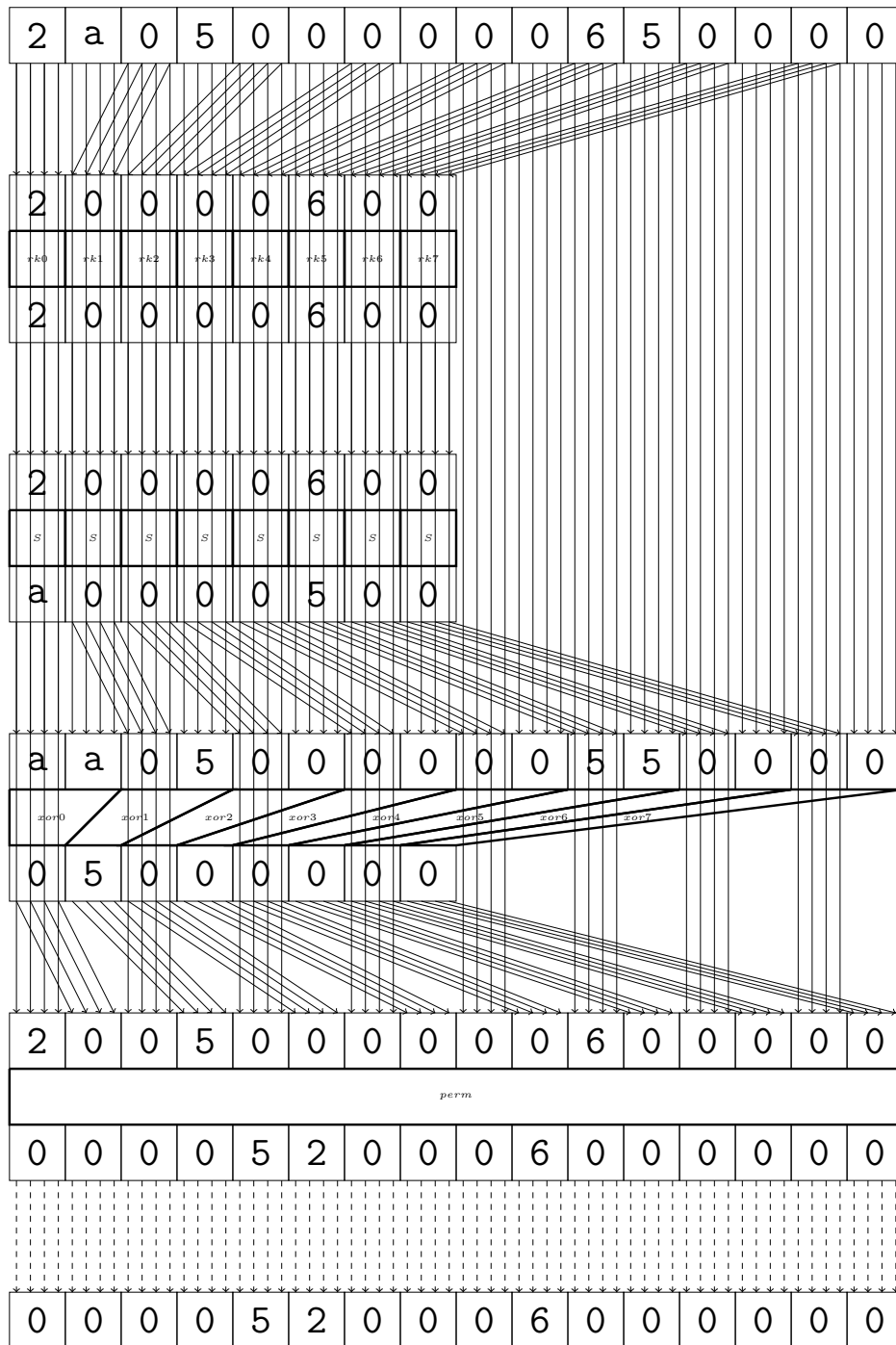
## 2 round

Maximal differential probability:  $2^{-6.0}$



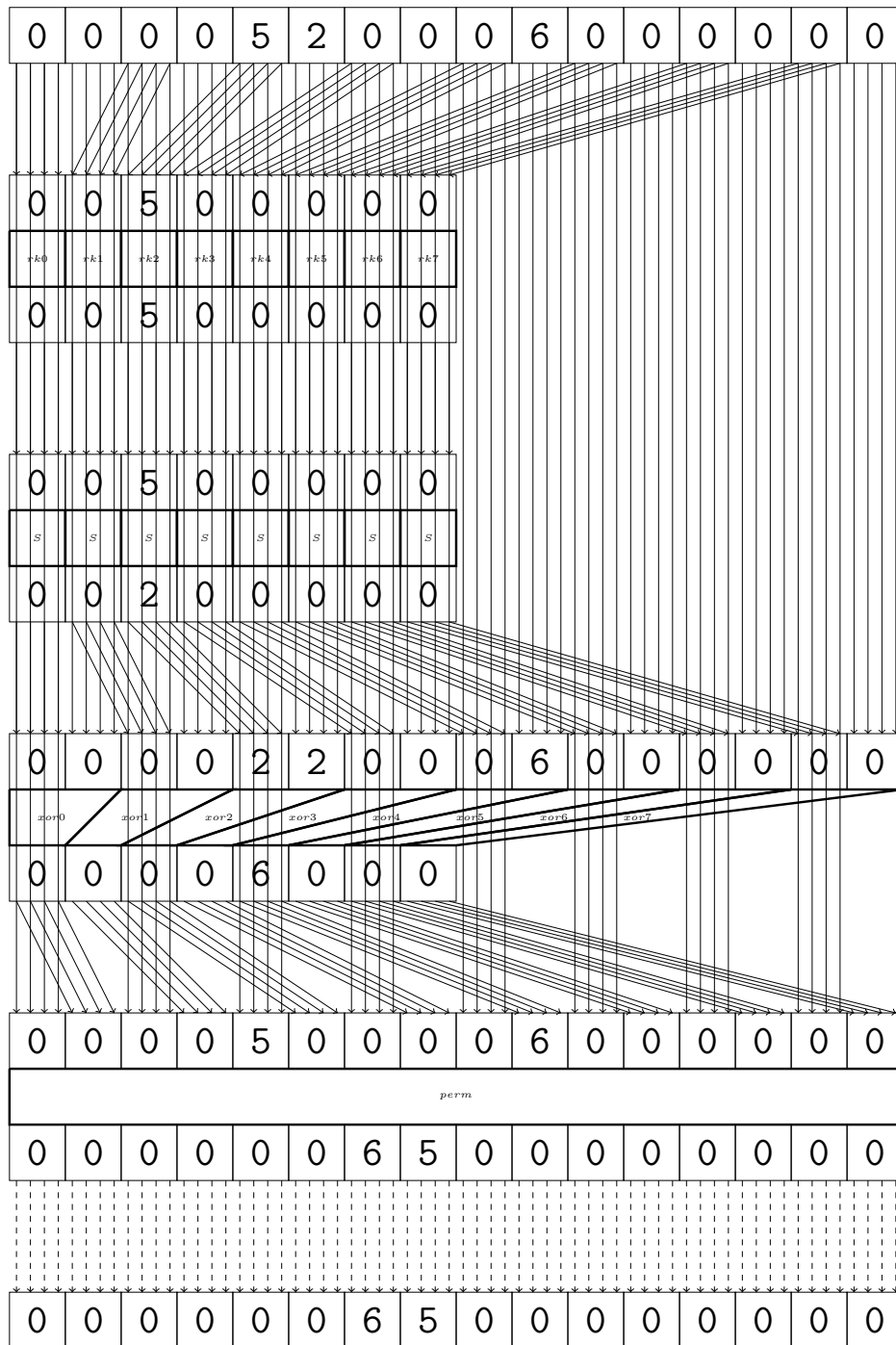
### 3 round

Maximal differential probability:  $2^{-4.0}$



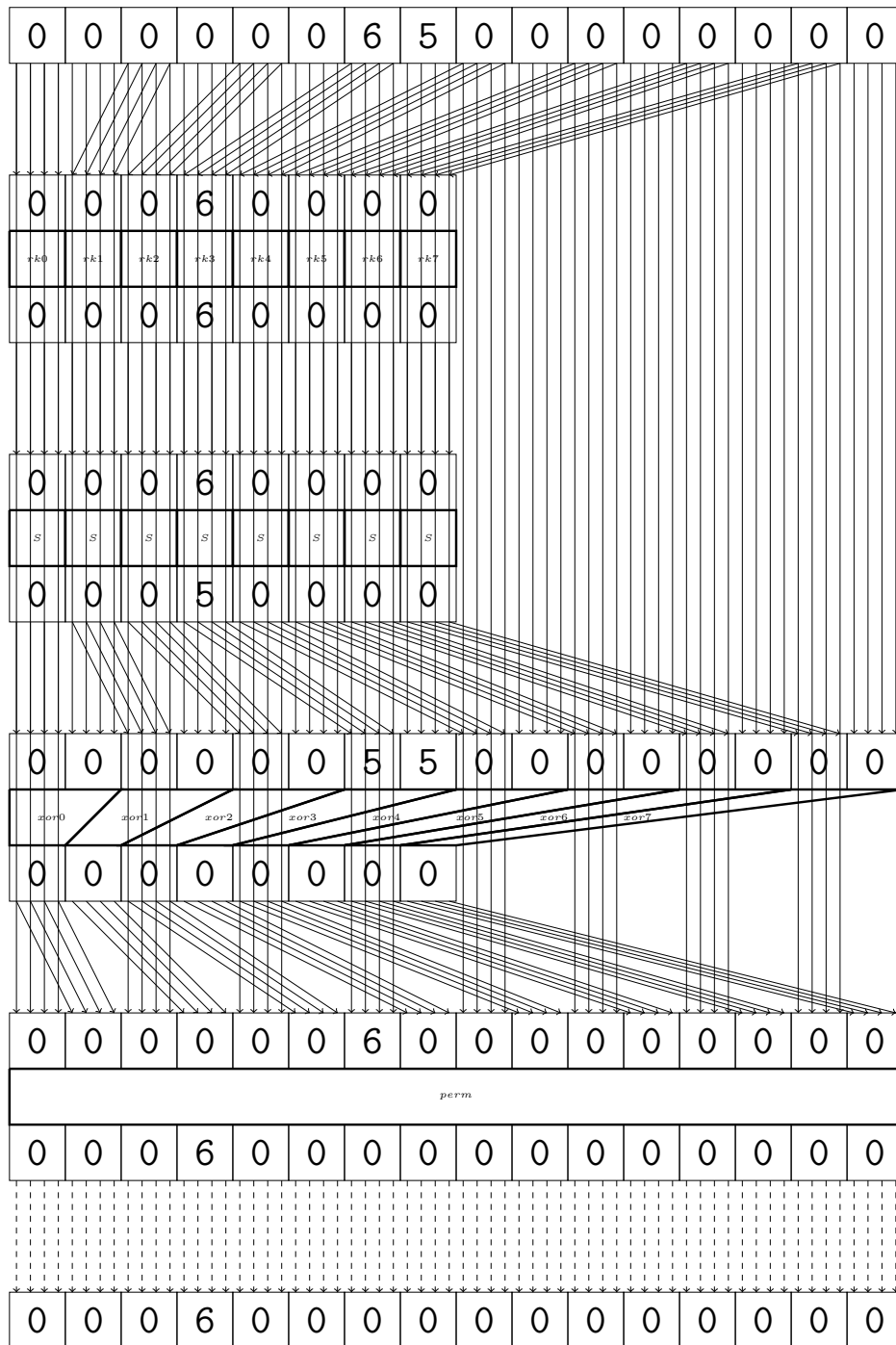
## 4 round

Maximal differential probability:  $2^{-2.0}$

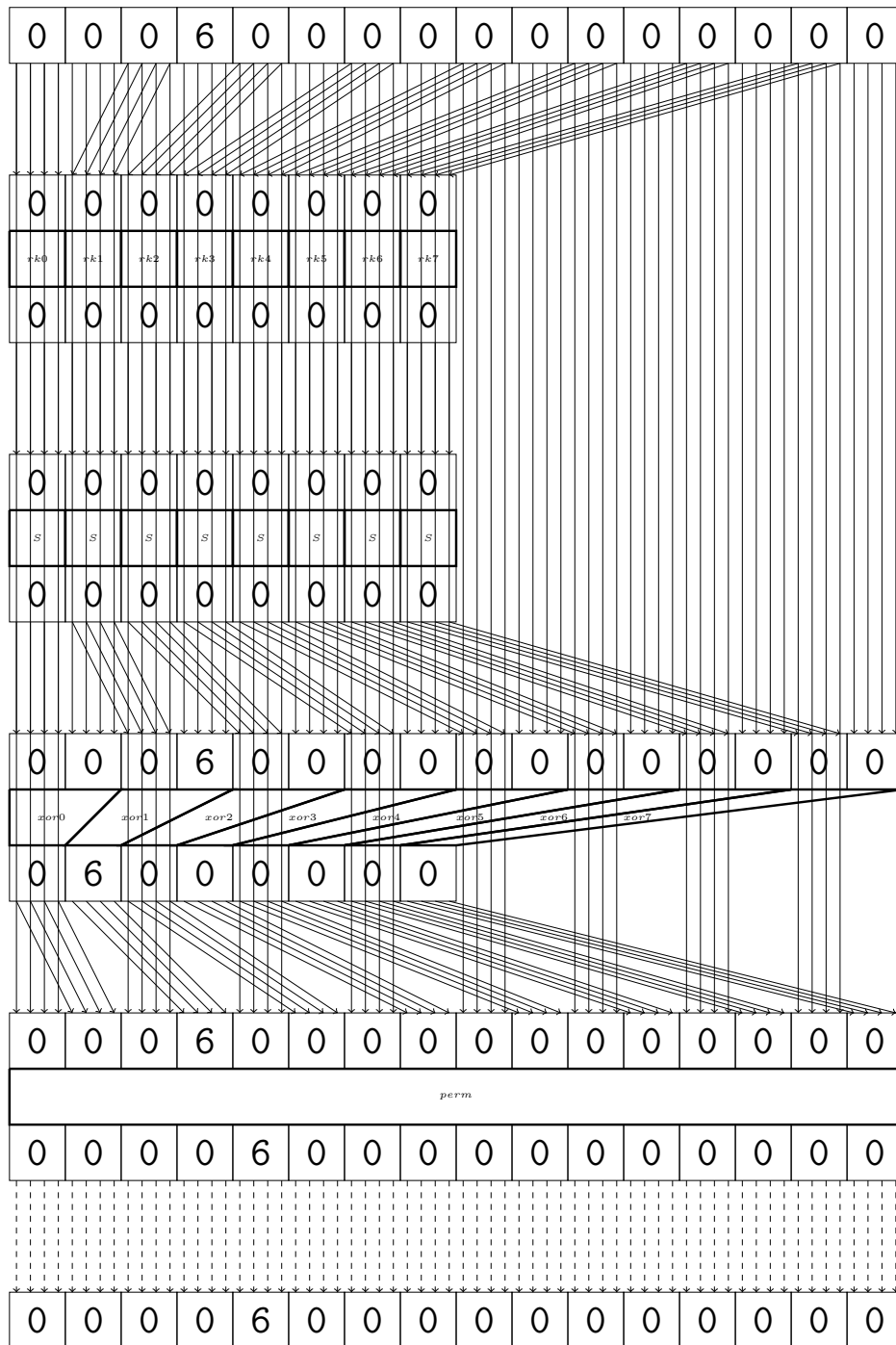


## 5 round

Maximal differential probability:  $2^{-2.0}$

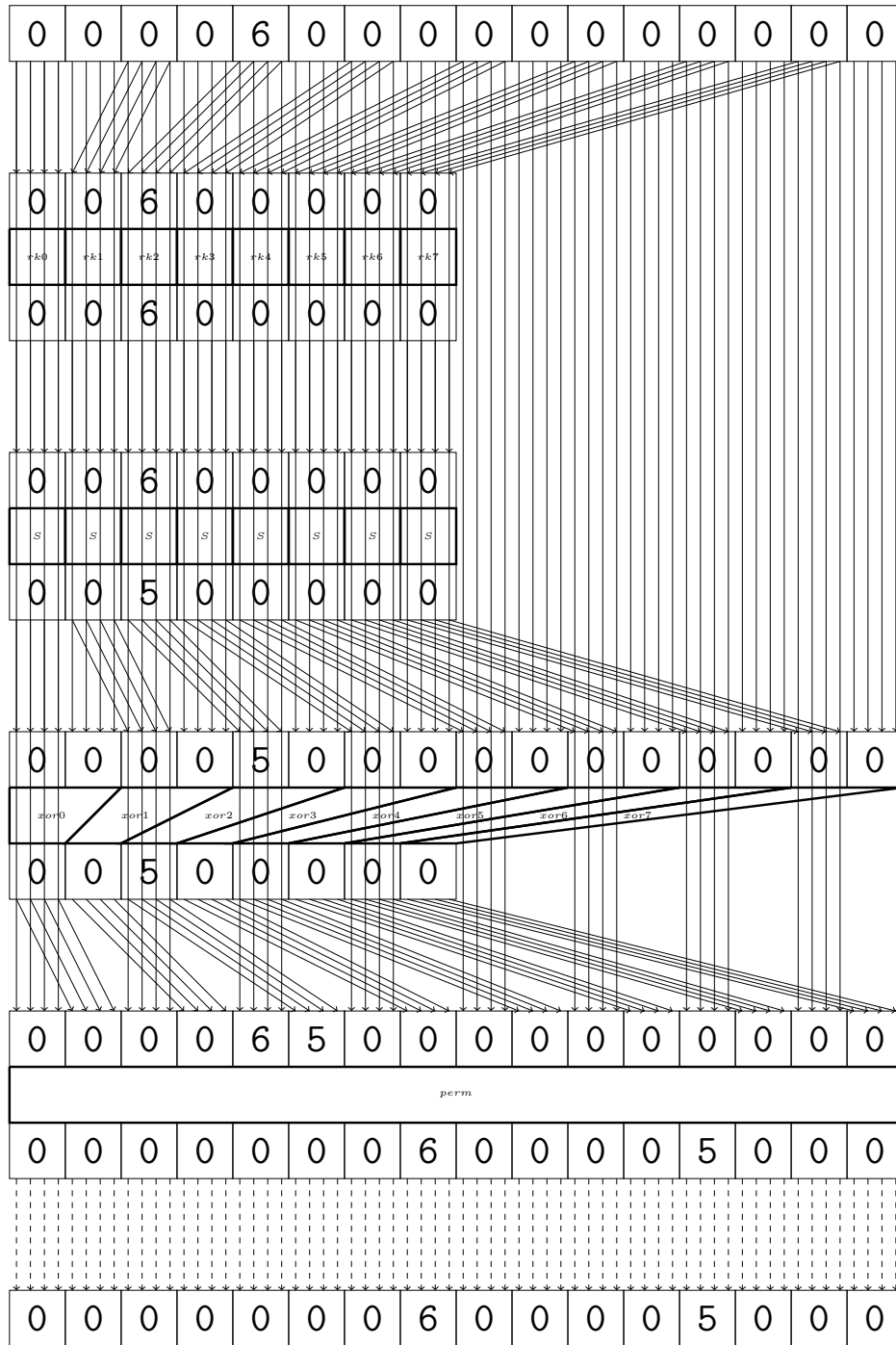


## 6 round

Maximal differential probability:  $2^{-0.0}$ 

## 7 round

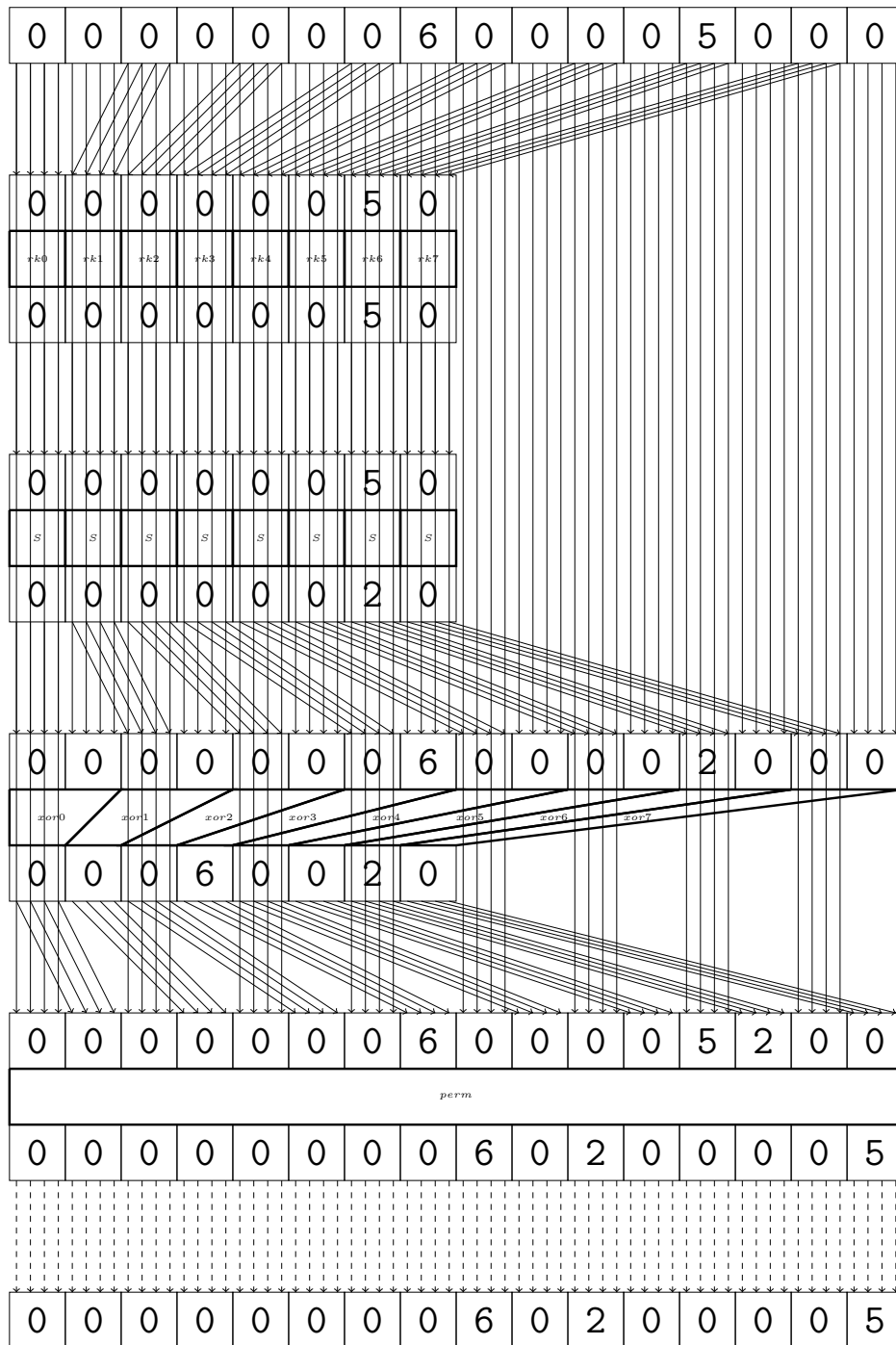
Maximal differential probability:  $2^{-2.0}$





## 8 round

Maximal differential probability:  $2^{-2.0}$



## 9 final

Maximal differential probability:  $2^{-4.0}$

